SAVER-VLA: Feedback-Driven Agentic Failure Recovery for Vision-Language-Action Manipulation

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Abstract: Vision-language-action (VLA) policies handle nominal manipulation well, but after a slip or displacement they keep executing actions that no longer apply. SAVER-VLA is an agentic layer that lets a pretrained VLA (SmolVLA) detect, diagnose, and recover from such failures without retraining. A verifier flags the failure, an LLM agent selects bounded recovery skills one at a time, and each measured outcome is fed back so the agent can replan or return control to the VLA. Across over 250 randomized LIBERO fault-injection episodes, SAVER-VLA raised task success from 3% to 66%; the LLM agent outperformed a fixed recovery script when recovery attempts failed (53% vs. 28%, paired $p = 0.006$), and ablations identify verified outcome feedback as the critical component.

Keywords: Agentic robotics, VLA models, Failure recovery, LLM agents

Project page: https://mehdiomid.github.io/saver-vla/

1 Introduction

VLA models map images and instructions directly to robot actions, yet they have no explicit notion of failure. A recent benchmark shows that state-of-the-art VLAs that succeed on nearly all nominal LIBERO tasks lose more than half their performance once a real execution failure occurs [1]. Language models can plan over skills, explain or correct failures [2–3], and decompose tasks for a VLA [4]. However, this reasoning is rarely used to supervise a pretrained VLA online, during its own execution.

We ask whether an LLM agent, invoked only after a detected failure, can guide a pretrained VLA through recovery: choosing bounded skills, interpreting their outcomes, and deciding when to hand control back. SAVER-VLA does this without retraining. We contribute (i) a training-free, event-triggered hierarchy in which an LLM agent supervises an unchanged VLA; (ii) verified outcome feedback that separates *target reached* from *grasp verified*, which our ablations show is critical; and (iii) a reproducible benchmark of randomized, paired fault injections that attributes success to the agent, the skills, or the policy.

2 Method: Detect, Diagnose, Act, Verify, Replan

Two loops at two time scales (Fig. 1a). SmolVLA (via LeRobot) drives a simulated Panda arm at 20 Hz. A verifier checks grasp and transport predicates every step; only on a failure is the agent called, and again after each skill or policy trial. The agent never outputs motor commands, keeping fast control and slow reasoning decoupled.

Agent interface. Each decision receives a JSON observation (grasp state, end-effector–object offset, last skill and trial outcomes, gripper-command mismatch, and available and blocked skills), never the fault's name. The agent (GPT-5.5) returns a skill, variant, and reason, so every action has a readable rationale.

Skills. Motion skills (pregrasp, approach, ±X grasp offset, 2D repositioning) issue bounded commands with timeouts and report `REACHED` or `FAILED`; reassess and abort complete the library.

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`RESUME_POLICY` starts a bounded SmolVLA trial that the verifier labels `VERIFIED` or `NOT_VERIFIED`: the agent positions the robot, and the learned policy performs the grasp.

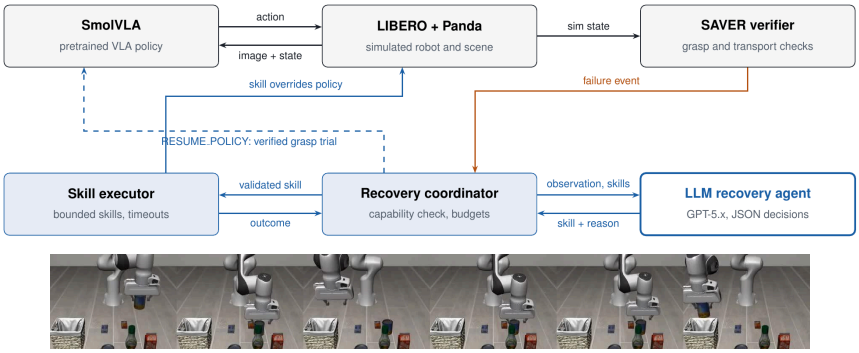


Figure 1: (a, top) Control loop and event-triggered recovery loop with outcome feedback. **(b, bottom)** After a centered grasp fails with no gripper mismatch, the agent applies a −X offset (5.9 → 1.3 cm) and the next trial is verified.

3 Experiments and Results

Setup. We evaluate `lerobot/smolvla_libero` on LIBERO-Object task 0 (soup can to basket) with privileged state over 256 episodes. Faults are randomized per episode (6–22 cm separation in a random direction, 10–40-step gripper release, sabotaged first handover, hidden ±5 cm approach bias) and seeded, so all controllers face identical faults. Initial states are ones SmolVLA solves without faults; the agent is credited only after a verified, agent-supervised grasp.

Experiment	Controller / variant	Task success (95% CI)
E1: randomized faults	SmolVLA alone	1/30 (3%; 1–17%)
2 fault families, 16 runs each	Restart policy (queue reset)	8/30 (27%; 14–44%)
	SAVER-VLA, scripted agent	19/30 (63%; 46–78%)
	SAVER-VLA, GPT-5.5	19/29 (66%; 47–80%)
E2: hard faults (paired)	Scripted agent	11/40 (28%)
5 scenarios × 8 initial states	GPT-5.5	21/40 (53%), p = 0.006
E3: ablations (paired, n = 8)	Full system	8/8
	Without verified grasp trials	1/8, p = 0.0014
	Without capability information	4/8
	Without geometry skills	4/8
	Fault disclosed to the agent	7/8
	Scripted agent	0/8, p = 0.0002

Table 1: Task success when the fault fired. Exact two-sided tests: E2 paired McNemar; E3 Fisher vs. full system.

Recovery is necessary. Under randomized faults, SAVER-VLA raised task success from 3% (SmolVLA alone) to 63–66%, well above restarting the policy (27%; paired $p \approx 0.004$). A fixed script matched the LLM agent on these common faults.

Reasoning helps when attempts fail. On hard faults, the LLM agent succeeded in 21/40 paired episodes versus 11/40 for the script (McNemar $p = 0.006$). After a sabotaged first grasp, the script aborted (0/8) while the agent diagnosed and retried (7/8): it retries the same grasp if the gripper ignored its commands, but shifts the grasp in −X if the hand sat 5.9 cm off target (Fig. 1b).

Outcome feedback is the critical component. Removing verified grasp trials, so the agent never learns whether a handover worked, cut success from 8/8 to 1/8 ($p = 0.0014$). Removing capability information or geometry skills each gave 4/8 (all differing episodes favored the full system; n.s. at $n = 8$). Disclosing the fault did not help (7/8): the agent diagnoses from evidence.

4 Limitations and Outlook

Results are simulated, with privileged state, on one task. Under combined faults neither agent succeeded (0/16): SmolVLA would not grasp objects displaced outside its training distribution, motivating agent-callable grasp skills. GPT-5.5 used 9–33k tokens and 19–58 s per episode.

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